

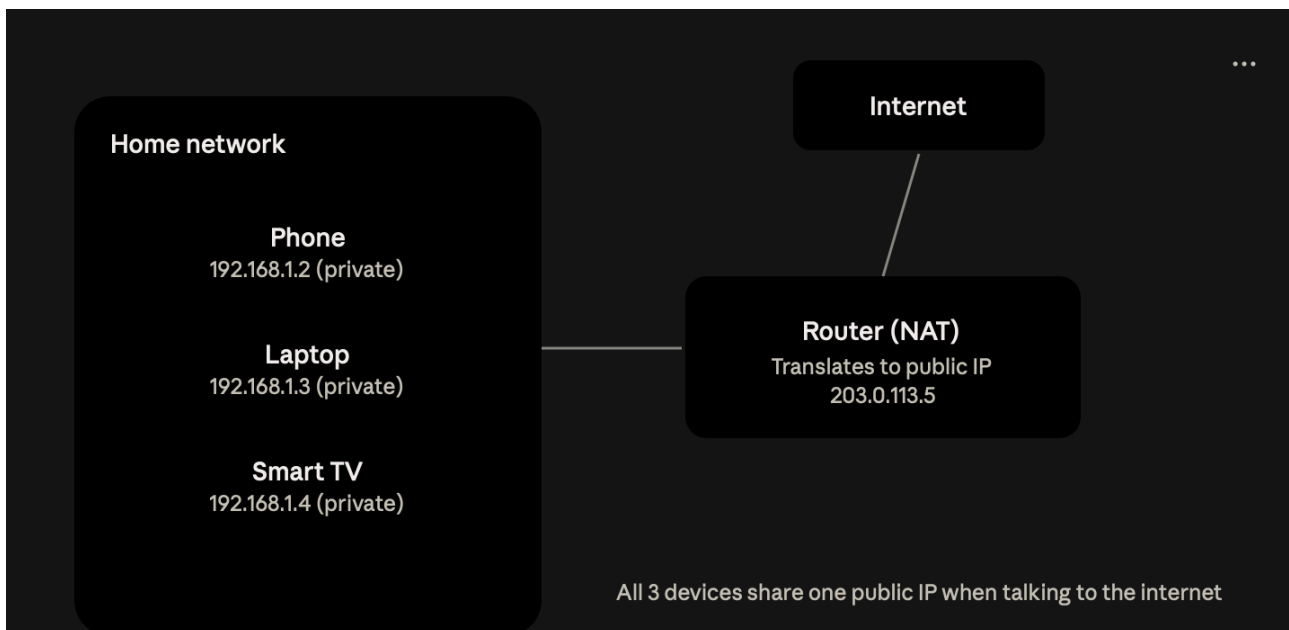
Module 2: Network + Network Security , Maintenance and Troubleshooting

Secure network with firewalls NAT, port filtering and other methods

Task 1:

Inbound Rules				
Name	Group	Profile	Enabled	Action
 Block Inbound Port 8080		All	Yes	Block

Task 2:



Task 3:

Blocking outgoing port 25 (SMTP) on your route

Why this helps prevent spam: Port 25 is the standard port for sending email (SMTP), so if malware infects a device on your network and tries to turn it into a spam-sending bot, blocking outbound port 25 stops it from directly relaying spam emails to the internet. This is also why most ISPs block port 25 by default on

residential connections — it's a common tactic against botnets that hijack home devices for mass spam campaigns.

Task 5:

How Instagram/Zomato might use firewalls and packet filtering

1. **Rate limiting and IP blocking at the edge** — filtering out IPs that send abnormally high request volumes (e.g., a bot scraping profiles or a DDoS attempt), before that traffic ever reaches the application servers.
2. **Restricting database access to internal IPs only** — their databases likely sit behind a firewall that only allows connections from the app's own backend servers, not from the public internet, so even if someone finds the database's address, they can't directly connect to it.
3. **Web Application Firewall (WAF) filtering malicious request patterns** — inspecting incoming HTTP requests for patterns like SQL injection or script injection attempts (e.g., someone trying to exploit a search bar or comment field) and blocking those requests before they reach the app logic.